

A Companion Study of Neural SDE Forecasting

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Abstract

This companion study extends the coordinate-wise robustness analysis of neural stochastic differential equation (SDE) forecasters from four named model families to a catalogue of 23 diffusion functions. We evaluate each diffusion on the 14-dimensional MuJoCo forecasting task with an input length of 50 and a forecast horizon of 20 and 40, and summarize every model by its horizon-dependent Monte Carlo predictive variance and mean-squared error (MSE). The resulting profiles form a continuum rather than a small set of sharply separated shape clusters. Nevertheless, two different associations are visible. Across the 23 single-diffusion architectures, mean predictive variance and mean MSE are strongly positively associated. Within an architecture, however, predictive variance typically increases with horizon while MSE first decreases and then rises; the median horizon-wise correlation is -0.61 . This distinction resolves the apparent contradiction between architecture-level and horizon-level conclusions.

We introduce a jointly trained three-branch root-mean-square (RMS) diffusion. Its nonnegative softmax gates mix component variances, while a learned global scale and bounded output gate control total stochastic intensity. Three preliminary component triplets are evaluated. The best mixture reaches mean MSE 0.0587 with mean predictive variance 6.19×10^{-4} , but it does not outperform the best single diffusion (MSE 0.0498).

Keywords: neural stochastic differential equations; time-series forecasting; diffusion function; predictive variance; mixture model; robustness

1 Introduction

Neural ordinary differential equations (ODEs), latent ODEs, and neural controlled differential equations (CDEs) provide continuous-time representations for irregularly sampled data and forecasting [1, 2, 3]. Neural SDEs add an explicit stochastic channel,

$$dz_t = f_\theta(t, z_t, X_t) dt + (g_\phi(t, z_t), dW_t, \quad (1)$$

so that one trained model defines a distribution of future paths rather than a single trajectory. Stable neural SDE constructions additionally emphasize that the form of the drift and diffusion can determine long-time behaviour [4]. The diffusion function g_ϕ is therefore not a minor implementation detail: it determines how stochastic perturbations enter the latent dynamics and, through the nonlinear drift and decoder, affects both the predictive mean and predictive dispersion.

Our preceding study, compared four representative neural SDE architectures with a neural CDE baseline [5]. That analysis showed that aggregate robustness can conceal coordinate-specific failure modes, and that high raw discrepancy and high normalized sensitivity need not occur in the same channel. The present paper asks a complementary question: what happens when the drift and evaluation protocol are held fixed while the diffusion function is varied systematically?

The initial hypothesis was that horizon-dependent variance profiles might separate into a few discrete classes—for example sublinear, approximately linear, and superlinear growth—and that a mixture of three representatives could adapt to a new data set. The experiment gives a more nuanced answer. The 23 functions span a largely continuous geometry, so a three-branch mixture is better understood as a compact adaptive basis than as a proven three-class taxonomy. Moreover, “variance of the data law” is not directly observed in this benchmark: the measured quantity is predictive variance induced by the learned diffusion, drift, decoder, numerical solver, and training procedure.

Table 1: The 23 single-diffusion candidates. “Lin” denotes one affine layer and “MLP” denotes a two-layer network with ReLU activation.

IDs	Family	Raw diffusion before the common output gate
g01–g03	scalar scale	$\sigma, \sigma t, \sigma z$
g04–g06	diagonal scale	$\sigma_j, \sigma_j t, \sigma_j z_j$
g07–g11	analytic state/time	$\sqrt{ z + \epsilon} - \sqrt{\epsilon}, z^3, \text{sigmoid}(z), \text{ReLU}(z), tz$
g12–g15	learned linear	$\text{Lin}(t), \text{Lin}(t)z, \text{Lin}(t, z), \text{Lin}(t, z)z$
g16–g19	learned MLP	$\text{MLP}(t), \text{MLP}(t)z, \text{MLP}(t, z), \text{MLP}(t, z)z$
g20–g21	nonlinear state	$\log(1 + z), \exp(\text{clip}(z))$
g22–g23	factorized learned	$\text{Lin}(t)\text{Lin}(z), \text{Lin}(t) + \text{Lin}(z)$

This paper makes four contributions:

1. it expands the MuJoCo diffusion sweep to 23 scalar, diagonal, state-dependent, time-dependent, linear, and MLP-based functions;
2. it separates absolute profile level from area-normalized profile shape, and separates between-architecture from within-horizon association;
3. it formulates and implements a jointly trained three-branch RMS diffusion with an interpretable local-covariance motivation; and
4. it reports preliminary mixture results and identifies the controls required before claiming general superiority.

2 Experimental Setting

2.1 Data and forecasting task

We use the 14-dimensional MuJoCo Hopper trajectories employed in the companion study. Each trajectory contains 100 time steps. An input window contains 50 steps and the target contains the following 20 steps. Sliding the window produces 3,000 strongly overlapping examples, corresponding to 2,100 training, 450 validation, and 450 test windows (70/15/15 source trajectories under the repository ordering). Because adjacent windows overlap heavily, uncertainty should ultimately be estimated at the source-trajectory level rather than by treating all windows as independent.

All SDE runs use the same latent-only drift option, hidden width 49, four hidden layers, Euler integration, training seed 12, and 200 requested epochs. The result artefacts analyzed here were produced with requested batch size 1,024; the legacy loader processed two full batches per training epoch and dropped the remaining 52 windows. Final SDE predictions use 20 Monte Carlo paths with seed 12,345. The stress grid contains missing rates $q \in \{0, 0.7\}$ and input-noise levels $\eta \in \{0, 0.1\}$ under random missingness, with five corruption repeats per cell. Profiles are averaged over these four cells.

This is a controlled catalogue comparison, not a complete hyperparameter search. In particular, the single training seed does not quantify optimization variability, and these stored results predate the revised fixed-panel Monte Carlo validation protocol in the accompanying code base.

2.2 Diffusion catalogue

Table 1 summarizes the 23 functions. The implementation uses diagonal Brownian noise. Scalar parameters are broadcast across latent coordinates, whereas diagonal parameters are coordinate-specific. Learned time features use $[\sin t, \cos t]$; state functions act on the latent state z . Every raw diffusion is subsequently gated and bounded by the benchmark’s outer nonlinearity.

2.3 Loss and variance profiles

For model m , forecast horizon h , example n , output coordinate j , and Monte Carlo path k , let $\hat{y}_{mnhj}^{(k)}$ be the prediction and let $\bar{y}_{mnhj} = K^{-1} \sum_k \hat{y}_{mnhj}^{(k)}$. We define

$$V_m(h) = \frac{1}{Nd} \sum_{n=1}^N \sum_{j=1}^d \text{Var}_k [\hat{y}_{mnhj}^{(k)}], \quad (2)$$

$$L_m(h) = \frac{1}{Nd} \sum_{n=1}^N \sum_{j=1}^d (\bar{y}_{mnhj} - y_{nhj})^2. \quad (3)$$

Thus $V_m(h)$ measures Monte Carlo predictive dispersion, whereas $L_m(h)$ measures the MSE of the Monte Carlo mean. Low $V_m(h)$ is not automatically better: a deterministic but biased model can have low variance and poor calibration. Conversely, MSE of the predictive mean does not reward a useful probabilistic spread.

To separate level from shape, we also use area normalization,

$$\tilde{V}_m(h) = \frac{V_m(h)}{\sum_{\tau=0}^{H-1} V_m(\tau)}, \quad \tilde{L}_m(h) = \frac{L_m(h)}{\sum_{\tau=0}^{H-1} L_m(\tau)}. \quad (4)$$

The loss–dispersion plane represents each model by $\bar{V}_m = H^{-1} \sum_h V_m(h)$ and $\bar{L}_m = H^{-1} \sum_h L_m(h)$.

3 A Learnable Three-Branch RMS Diffusion

3.1 Why diffusion components are combined in variance space

Consider the Itô SDE

$$dz_t = f(t, z_t) dt + G(t, z_t) dW_t. \quad (5)$$

Its integral form separates the drift contribution from the stochastic integral,

$$z_t = z_0 + \underbrace{\int_0^t f(s, z_s) ds}_{A_t} + \underbrace{\int_0^t G(s, z_s) dW_s}_{M_t}. \quad (6)$$

For a predictable square-integrable diffusion matrix G , the Itô integral has zero mean and the matrix-valued Itô isometry gives

$$\text{Cov}(M_t) = \mathbb{E} \left[\int_0^t G(s, z_s) G(s, z_s)^\top ds \right]. \quad (7)$$

Equivalently, the instantaneous covariance rate contributed by the diffusion is $G(t, z_t) G(t, z_t)^\top$. This is the sense in which the stochastic variance depends quadratically, rather than linearly, on the diffusion amplitude. Equation (7) describes the martingale part, not the complete predictive covariance of z_t . In a neural SDE the drift is state dependent, so A_t is random and generally correlated with M_t :

$$\text{Cov}(z_t) = \text{Cov}(z_0 + A_t) + \text{Cov}(M_t) + \text{Cov}(z_0 + A_t, M_t) + \text{Cov}(M_t, z_0 + A_t) = \text{Cov}(A_t) + \text{Cov}(M_t) + 2 \text{Cov}(A_t, M_t) \quad (8)$$

Consequently, matching the local diffusion covariance does not imply that the final horizon-dependent predictive variance is an exact linear mixture after nonlinear drift propagation and decoding.

Now let three diffusion matrices share the same Brownian increment and define $G_{\text{sum}} = \sum_{i=1}^3 a_i G_i$. Substitution into the covariance rate yields

$$G_{\text{sum}} G_{\text{sum}}^\top = \sum_i a_i^2 G_i G_i^\top + \sum_{i < j} a_i a_j (G_i G_j^\top + G_j G_i^\top). \quad (9)$$

The cross terms arise because all components respond to the same realization of dW_t ; depending on their signs and directions, they can reinforce or cancel one another. Therefore, a linear sum of amplitudes is not

a convex mixture of their variance contributions. If each component is instead driven by an independent Brownian motion, then the covariance rate becomes $\sum_i a_i^2 G_i G_i^\top$, but the Brownian dimension and simulation cost increase with the number of branches.

For the diagonal-noise model used here, the same local covariance mixture can be represented with a Brownian motion of the original dimension. Write $G_i(t, z) = \text{Diag}(\tilde{g}_i(t, z))$, where $\tilde{g}_i$ is the vector of raw component amplitudes. We introduce nonnegative normalized weights and a separate global scale,

$$\pi_i = \frac{\exp(\alpha_i)}{\sum_{r=1}^3 \exp(\alpha_r)}, \quad \sum_i \pi_i = 1, \quad \pi_i \geq 0, \quad (10)$$

$$s = \exp(\text{clip}(\ell_s, -20, 10)), \quad (11)$$

and define the effective pre-cap diffusion coordinate-wise as

$$g_{\text{mix}}(t, z) = s \sqrt{\sum_{i=1}^3 \pi_i \tilde{g}_i(t, z)^2 + \varepsilon}, \quad G_{\text{mix}} = \text{Diag}(g_{\text{mix}}). \quad (12)$$

The square and square root act element-wise. Ignoring the small numerical term ε , this construction satisfies

$$G_{\text{mix}} G_{\text{mix}}^\top = s^2 \sum_{i=1}^3 \pi_i G_i G_i^\top. \quad (13)$$

Thus the learnable gates interpolate between the components' *local variance contributions*, while s controls their total magnitude. The identity is exact before the implementation's final bounded stability cap; it becomes approximate when that cap operates outside its near-linear regime.

3.2 Implemented effective diffusion

The current benchmark implementation uses

$$g_{\text{eff}}(t, z) = \tanh \left(\text{sigmoid}(\theta) s \sqrt{\sum_{i=1}^3 \pi_i \text{clip}(\tilde{g}_i(t, z))^2 + \varepsilon} \right). \quad (14)$$

Here α_i are learnable gate logits, π_i are nonnegative and sum to one, ℓ_s is a learnable log-scale, θ is the benchmark's common output gate, and $\varepsilon = 10^{-12}$ prevents an undefined square root and zero-gradient singularity. Components are clipped before squaring to prevent overflow. The outer tanh bounds the numerical amplitude, but it also means that the exact covariance-mixture interpretation holds before saturation and only approximately in the linear region of the cap.

Each branch has its own parameters, but all branches, the drift, and the forecasting head are trained jointly. We do not add checkpoints from separately trained models because their latent coordinates and drifts need not be compatible. The first experiment uses global gates; conditioning the gates on the observed history is left for future work because it introduces a more powerful model and a greater risk of leakage or overfitting.

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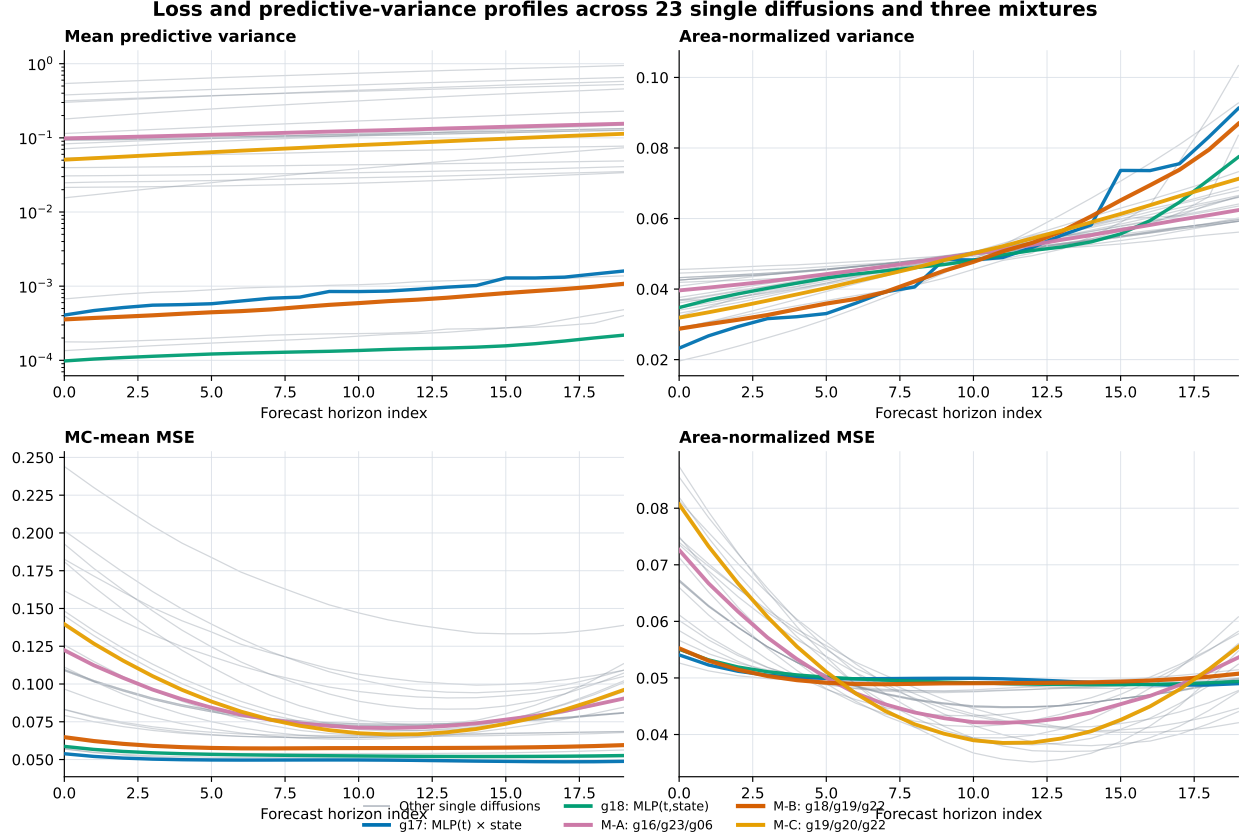


Figure 1: Absolute and area-normalized predictive-variance and loss profiles. Thin gray curves denote the remaining 21 single diffusions; g17 and g18 are highlighted together with the three RMS mixtures. Absolute variance is plotted logarithmically because its scale spans several orders of magnitude.

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4 Results

4.1 Profiles form a continuum

Figure 1 shows absolute and area-normalized profiles. The absolute variance spans more than three orders of magnitude. State-amplifying functions such as g03, g06, and g07 occupy the high-dispersion region, whereas several learned functions (g16–g19 and g22) learn an almost deterministic regime. Area normalization compresses the scale differences and exposes a continuous range of growth shapes. The curves do not justify a hard claim that exactly three natural clusters exist.

Most variance profiles grow with horizon, while most loss profiles are U-shaped: loss decreases over the early and middle forecast indices and rises again near the end. Across the 23 single diffusions, every horizon-wise correlation between $V_m(h)$ and $L_m(h)$ is negative; the median is -0.61 and the range is $[-0.91, -0.15]$. This does not mean that more diffusion improves forecasting. Horizon is a common confounder: the latent readout can fit middle target indices better even while Brownian uncertainty accumulates.

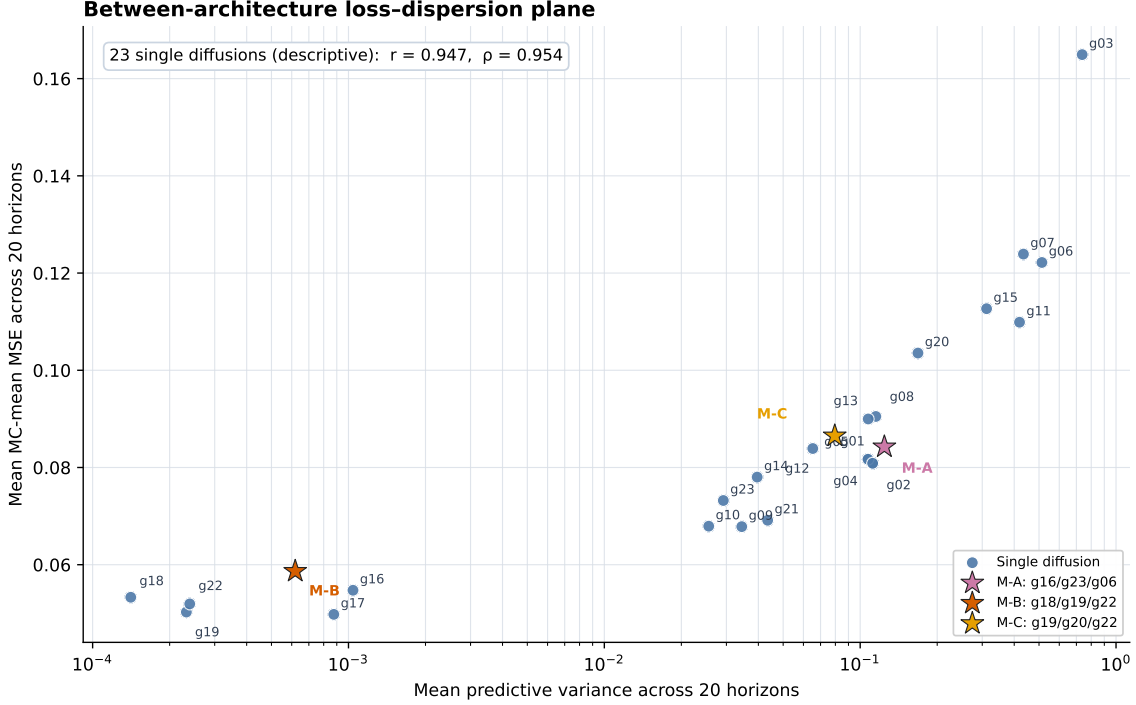


Figure 2: Mean predictive variance versus mean MC-mean MSE across the 20-step horizon. The horizontal axis is logarithmic. Correlations are descriptive statistics for the 23 single-diffusion design points; the three mixtures are shown as stars and are not included in the reported correlations.

4.2 Between-architecture association is strong

The architecture-level picture is different. Figure 2 places each model at $(\bar{V}_m, \bar{L}_m)$. Among the 23 single diffusions, the descriptive Pearson correlation is 0.947 and the rank correlation is 0.954. Architectures with larger average predictive variance therefore also tend to have larger average MC-mean MSE in this fixed MuJoCo protocol.

This association is descriptive rather than causal. The 23 points are designed architectures, not independent random samples; they differ in parameterization and optimization geometry. In addition, mean predictive variance is not a property of the data set alone. It is an output of the jointly trained SDE and depends on the drift, decoder, solver, objective, and random seed. The result supports screening diffusion functions by a measured loss–dispersion trade-off, but it does not support selecting a diffusion from an independently estimated scalar “data variance.”

4.3 Preliminary mixture experiment

Table 2 reports the three component triplets. The best mixture, M-B, combines g18, g19, and g22. It achieves mean loss 0.0587 and mean variance 6.19×10^{-4} . Relative to the best-loss single diffusion g17, M-B reduces variance by approximately 29% but increases MSE by approximately 18%. It is also dominated by g18 in the two plotted metrics: g18 has both lower mean variance and lower loss. The mixture therefore demonstrates a working implementation and a tunable profile, but not a new state of the art.

The broad spread among M-A, M-B, and M-C shows that component selection remains important. Global gates cannot guarantee that a weak component set becomes a strong predictor, and the final tanh cap can partially collapse differences between large raw amplitudes. A fair next comparison should include a single wider diffusion network with the same parameter count as the mixture.

Table 2: Mean metrics across 20 horizons and four stress conditions. Lower is better for the two reported quantities, but predictive variance alone is not a calibration score.

Model	Components	Parameters	$\bar{V}$	$\bar{L}$
g17	$\text{MLP}(t) \times z$	19,468	8.76×10^{-4}	0.0498
g18	$\text{MLP}(t, z)$	21,869	1.41×10^{-4}	0.0533
M-A	g16/g23/g06	22,118	1.24×10^{-1}	0.0843
M-B	g18/g19/g22	29,468	6.19×10^{-4}	0.0587
M-C	g19/g20/g22	24,470	7.96×10^{-2}	0.0865

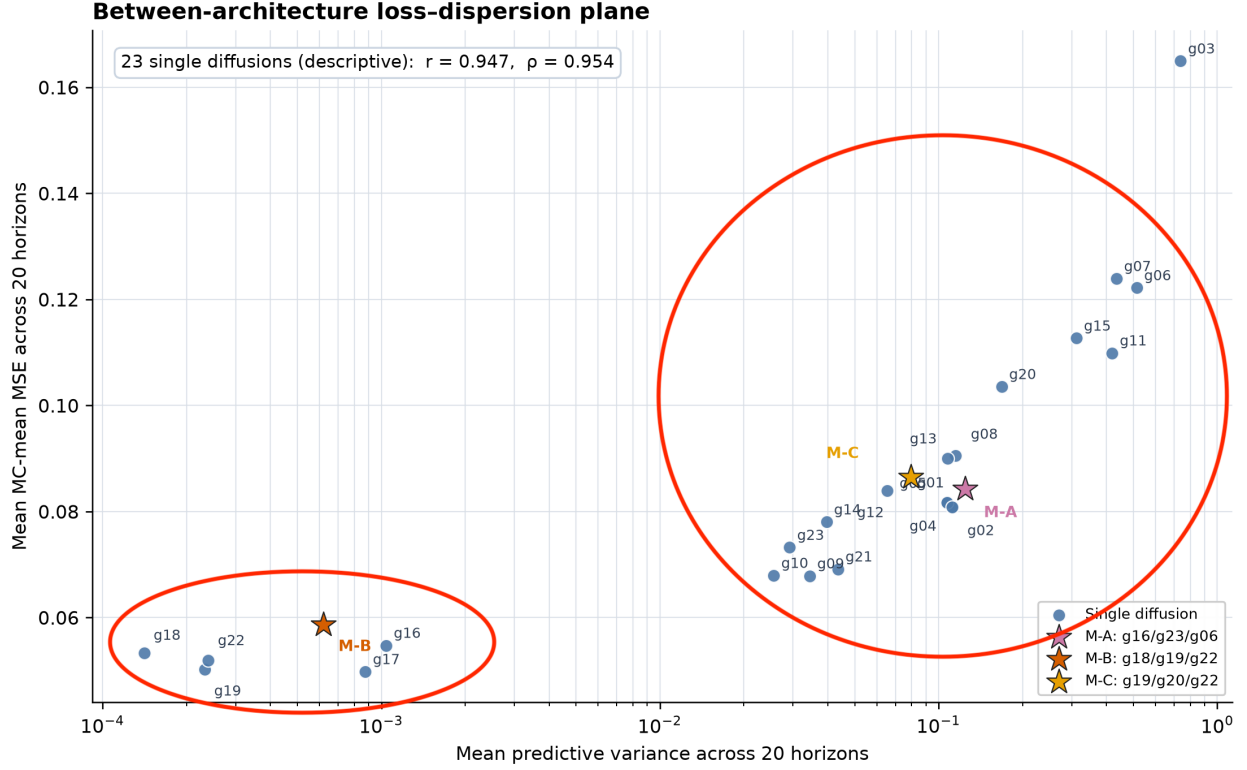


Figure 3: Overall clusters

4.4 Dimension-wise profiles

As reported in [5], the MuJoCo coordinates have substantially different noise profiles. In particular, coordinates d_{10} and d_{11} exhibit larger normalized predictive variance than most of the remaining coordinates. Figures 4–7 therefore compare the architecture-level loss–dispersion plane separately for all 14 output coordinates. As we can see for all of those dimensions we have 2 clusters as showed above. The position of this cluster can change from dimension to dimension, however the function in it remaining almost unchanged. However for dimension 4 we are having huge gap between our model loss and the function which are mixed in it(m26). The reason of this gap remains unknown yet and worth a research

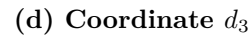
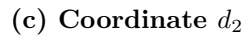
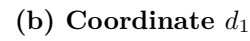
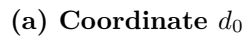
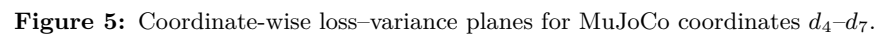
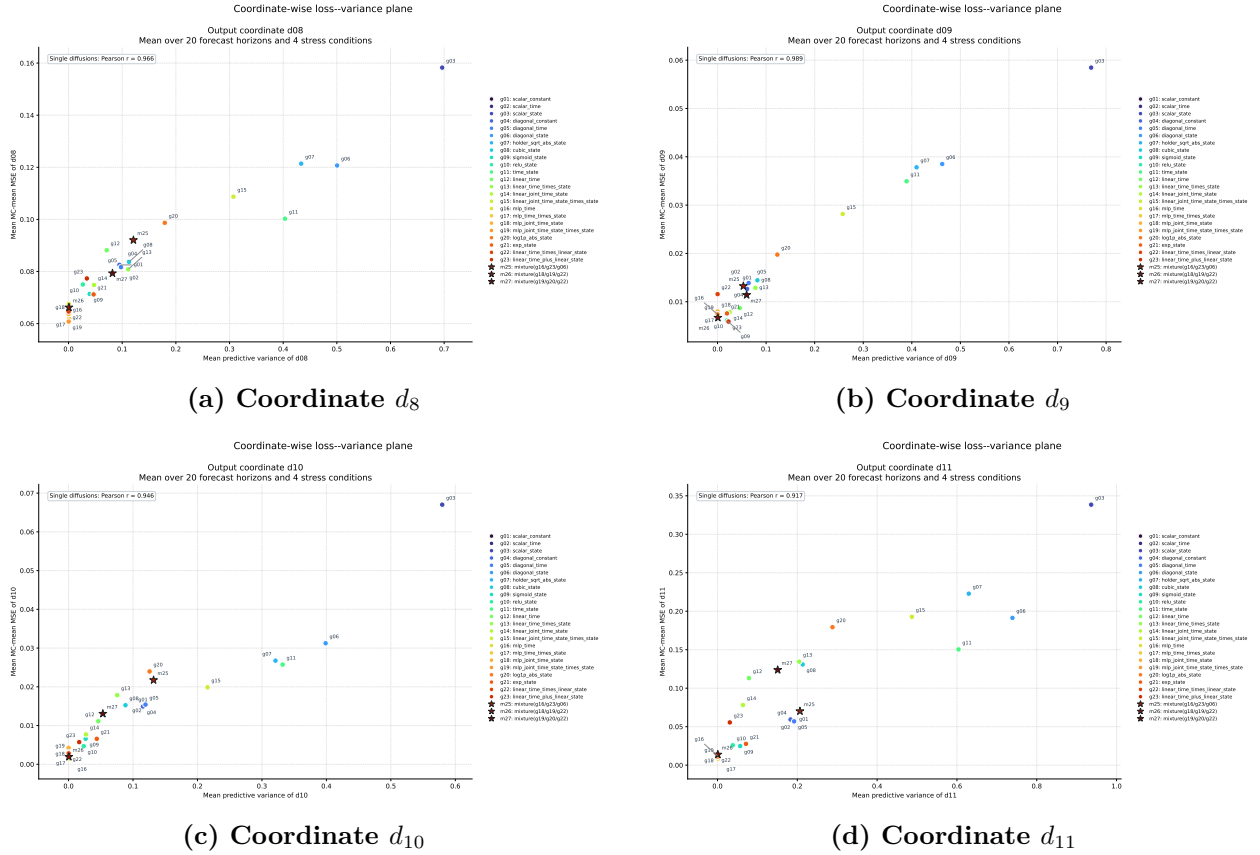


Figure 4: Coordinate-wise loss-variance planes for MuJoCo coordinates d_0 - d_3 .



Figure 6: Coordinate-wise loss-variance planes for MuJoCo coordinates d_8 – d_{11} .

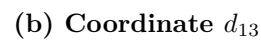


Figure 7: Coordinate-wise loss–variance planes for MuJoCo coordinates d_{12} and d_{13} .

5 Discussion

Between-model and within-model effects must not be conflated. The positive architecture-level association and negative horizon-level association answer different questions. The former asks which diffusion parameterizations produce low average dispersion and loss. The latter asks how two profiles evolve as the forecast index changes inside one trained model. Reporting only one of these summaries can lead to apparently contradictory conclusions.

Area normalization describes shape, not performance. Two models can have nearly identical normalized profiles and differ by orders of magnitude in absolute variance. Conversely, a model with an unusual normalized curve can still have negligible absolute stochasticity. Model selection should therefore retain both absolute and normalized views.

Predictive variance is not calibration. The present loss evaluates the Monte Carlo mean. It can favour a model whose diffusion collapses toward zero. A probabilistic forecast should additionally be evaluated by coverage, interval width, calibration error, negative log-likelihood when a density is available, or a proper sample-based score. Any conclusion about “optimal stability” without these checks would be premature.

The mixture is identifiable only with constraints. Softmax gates enforce $\pi_i \geq 0$ and $\sum_i \pi_i = 1$, while the separate scale s distinguishes total intensity from relative branch contribution. Even with these constraints, rescaling internal component networks can trade off against s and θ . Logging learned gates, scale, effective diffusion norms, and saturation frequency is necessary for interpretation.

Generalization requires a revised protocol. The next experiment should train at least five seeds under identical data, update budget, and fixed-panel validation; reevaluate every checkpoint with an independent larger Monte Carlo panel; and aggregate uncertainty by source trajectory. Comparisons should include (i) the best single diffusion, (ii) the three-branch RMS mixture, (iii) a shared-noise linear sum as an ablation, and (iv) a parameter-matched wider single branch. PhysioNet is a particularly useful next data set because its 41-channel input and 36-channel forecast target introduce genuine clinical missingness and a different sampling structure. Also Cryptocurrency Historical data will be researched since it is the data with very high inner data noise. The mixture should not be made history-conditional until the global-gate baseline is stable and free of future leakage.

6 Limitations

This study has five principal limitations. First, the reported sweep uses one training seed, so the observed ordering may include initialization and optimization effects. Second, the stored experiment uses legacy batch and validation semantics; new runs should use the revised non-dropping loaders and fixed Monte Carlo validation panel. Third, the 23 candidates do not have equal parameter counts, and three-branch mixtures are larger than most individual functions. Fourth, the stress grid contains four conditions and does not span block or tail missingness, stronger input noise, or distribution shift. Finally, neither MSE nor Monte Carlo variance establishes probabilistic calibration. The numerical correlations in this paper are therefore descriptive properties of one benchmark protocol, not universal constants.

7 Conclusion

Expanding the diffusion sweep from four named neural SDE families to 23 functions reveals a continuous loss–variance geometry. On MuJoCo, average predictive variance and average MC-mean MSE are strongly associated across architectures, while their horizon profiles are typically negatively associated within a model. A learnable three-branch RMS diffusion provides a mathematically cleaner alternative to summing amplitudes and can reproduce a range of profiles without increasing the Brownian dimension. The first three mixtures, however, do not surpass the best single diffusion. The appropriate conclusion is therefore

constructive but limited: the mixture is a viable model class that now requires multi-seed, parameter-matched, calibration-aware testing on MuJoCo.

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